

a dedicated graphical language and a simple graphical user interface (GUI) to facilitate non-programming users.

**Portability.** The software gives a lot of freedom in the choice of hardware and software for BCI. The user can go for different data-acquisition machines [such as an EEG or magnet-encephalography (MEG)], different software platforms (Windows or Linux for both 32-bit and 64-bit architectures using either Visual Studio 2010 or 2008 or GCC) and different visualisation techniques.

**Bundled software packages.** The environment is composed of many free and portable software including Gnome ToolKit (GTK), IT++ , GNU Scientific Library (GSL), Virtual Reality Peripheral Network (VRPN) and GNU Compiler Collection (GCC).

GTK software helps the user to have a fine graphical interface. It offers cross-platform compatibility and an easy-to-use application program interface. IT++ is a C++ library of mathematical, signal processing and communication routines.

GNU Scientific Library is yet another numerical library for C and C++ programmers, while VRPN is a set of classes within a library that helps in implementing an interface between application programs and the set of physical devices used in a VR system. GCC is a compiler that supports a wide range of architectures.

#### Scripting using Lua language.

Lua powers the scripting in this environment. It can be used to script scenarios and control behaviours. A special Lua Interpreter Box is readily available in OpenViBE that can handle stimulations.

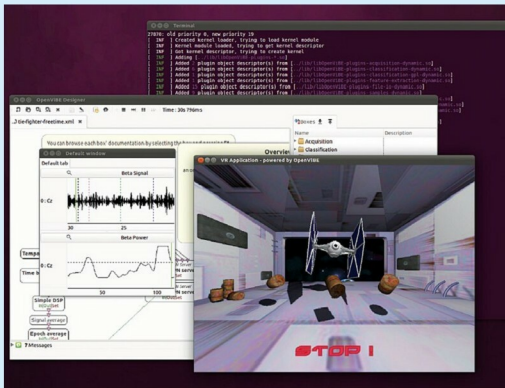


Fig. 3: OpenViBE facilitates communication with virtual reality applications

## The virtual museum application

This is a BCI application that helps a well-trained user to move around in a virtual museum by three mental commands: imagined movements of the right hand, left hand and the feet. The right hand movement allows a completely emotionless user to move right and the left hand movement to move left. The imagined foot movement allows the user to move forward. Such an application can only be used for a short duration as the intense concentration required to move around rapidly exhausts the user.

For quick navigation of the user to destination, the application has incorporated a new approach. Introduction of intermediate navigation points allows the user to access these points using the imagination of left or right hand movements. Once these points are selected, negotiation to these points becomes automatic. Imagination of feet movement could be used to quit or cancel selection.

Besides this, developers can also make use of Python Scripting. They can leverage the power of SciPy and NumPy libraries using Python.

## How is this package different

Most BCI software demand some level of programming knowledge from users. The smart graphical interface in this environment eliminates coding literacy from non-programming users. The range of possible designs is limited in most BCI software due to coarser modularity. Obviously, there are exceptions as in the case of BioSig software.

Again, OpenViBE environment is based on free and open source software. BioSig is completely open source but rtsBCI package needed for online and real-time BCI requires MATLAB/Simulink, which is non-free and proprietary software.

The software makes use of offline analysis for creating online scenarios. Embedded tools for innovative VR displays as well as real-time 3D visualisation of brain activities make OpenViBE an ideal choice for any VR application. ●

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